ACTUARIAL PROJECT · INDIAN MARKET · FY2025

Ind AS 117 / IFRS 17 · IRDAI FY27 deadline

Life Insurance CSM Model *under Ind AS 117*

1,000 simulated Indian term life policies · IALM 2012-14 · RBI G-Sec · Monte Carlo Risk Adjustment · Python 3

BEST ESTIMATE LIABILITY	RISK ADJUSTMENT	CSM AT INCEPTION	TOTAL ICL DAY-1
−₹1.607	+₹0.997	+₹2.345	₹1.735
crore · profitable book	crore · 75th pctile VaR	crore · deferred profit	crore · balance sheet

CSM ROLL-FORWARD — 20 YEARS · LOCKED-IN RATE 6.53%

Rising	Declining	Zero
Yrs 1–4	Yrs 5–12	Yr 20

MAJOR ASSUMPTIONS

Mortality basis	IALM 2012-14 (IAI)
Female mortality	78% of male (LIC L-42)
Discount curve	RBI G-Sec 6.40%-7.00%
Expense ratio	15.6% · IRDAI AR 2024-25
Age distribution	Uniform 25-55 · avg 40.1 yrs
Gender mix	70% male / 30% female
Sum assured	Log-normal · avg ₹27.7L
Claim timing	Mid-year UDD · v(t−0.5)
Monte Carlo	500 scenarios · rbinom

KEY INSIGHTS

Negative BEL is correct
Future premium inflows exceed future claims + expenses in PV terms. Profitable book. Surplus becomes CSM.

CSM prevents day-one profit
₹2.345 cr deferred. Released at ₹0.131–0.389 cr/yr as service delivered. Zero at year 20.

CSM rises in years 1–4
Interest accretion (6.53%) exceeds early releases. Policyholders aged ~41 face very low mortality (qx ≈ 0.002).

Monte Carlo quantifies RA
Claims PV ranged ₹11.5–15.5 cr across 500 scenarios. RA = 75th pctile − mean. Cannot be done in Excel.

Age drives everything
Age 25: net prem ≈ ₹3,100. Age 50: ≈ ₹22,000 (same ₹25L SA). Older entries may be onerous.

Gender mix affects profit
Female mortality 78% of male → cheaper to insure. 50/50 split would make BEL ₹0.14 cr more negative.

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IALM 2012-14 · IAI | CCIL G-Sec Yields | RBI DBIE
IRDAI Annual Report 2024-25 | Python 3 · numpy · scipy